SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Project Initialization and Planning Phase

Date	10 Jul 2026
Team ID	
Project Title	Smart Lender – AI-Powered Loan Approval Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed Smart Lender AI system aims to improve the loan eligibility assessment process using machine learning techniques. The system assists bank employees by providing accurate loan eligibility predictions, reducing manual effort, minimizing decision-making errors, and improving operational efficiency. Key features include AI-based loan prediction, applicant profile management, prediction history, eligibility analysis, and PDF report generation.

Project Overview	
Objective	The primary objective of the Smart Lender AI project is to develop an AI-based loan eligibility prediction system that enables bank employees to evaluate loan applications quickly, accurately, and efficiently using machine learning.
Scope	The project focuses on predicting loan eligibility based on applicant information, maintaining applicant profiles and prediction history, generating PDF reports, and assisting bank employees in making informed loan approval decisions.
Problem Statement	
Description	Traditional loan approval processes are time-consuming, require extensive manual verification, and may lead to inconsistent decision-making. Evaluating multiple applicant records manually increases workload and delays loan approvals.

Impact	These challenges reduce operational efficiency, increase processing time, and may result in inaccurate loan decisions. An intelligent loan prediction system helps improve accuracy, speed, and overall customer satisfaction.
Proposed Solution	
Approach	The proposed solution uses machine learning algorithms to analyze applicant information and predict loan eligibility. The system also maintains applicant profiles, stores prediction history, and generates detailed reports to support loan approval decisions.
Key Features	AI-Based Loan Eligibility Prediction Applicant Profile Management Prediction History Tracking Eligibility Score & Risk Analysis PDF Report Generation EMI Calculator AI Recommendation & Decision Support Responsive Web Interface

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU specifications for model training and application execution	Intel Core i5 / AMD Ryzen 5 or higher
Memory	RAM required for development and testing	8 GB
Storage	Space for datasets, trained models and project files	512 GB SSD
Software		

Frameworks	Backend Framework	Flask
Libraries	Data preprocessing and prediction	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Visual Studio Code (VS Code
Data		
Data	Source, size, format	**Kaggle Loan Approval Prediction Dataset, 4,269 applicant records, CSV file (.csv)**